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Hypophosphite Is a Naturally Occurring Selective Inhibitor of Syntrophic Methanogenesis

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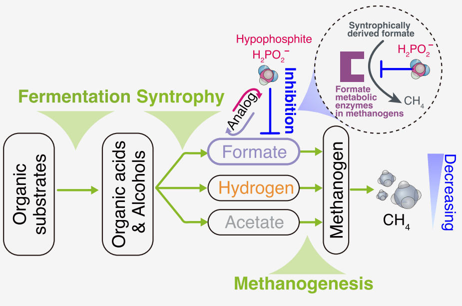
ABSTRACT: Microbial methanogenesis is a major contributor to global warming, and methane fluxes represent a loss of energy and electrons from industrial ecosystems. The chemical space of methane control strategies is still underexplored. Most known methanogenesis inhibitors target methanogenic archaeal enzymes. However, interference with the exchange of syntrophic electron carriers (H_2 or formate) in methanogenic systems presents an additional target for methane control. Here, we show that hypophosphite ($H_2PO_2^-$), an inorganic formate analogue, is a potent and selective inhibitor of syntrophic methanogenesis versus primary fermentation in rice field sediments and cattle rumens. Hypophosphite is also generally recognized as safe and relatively nontoxic to plants and animals. Genetic screens and physiological assays in the model methanogen *Methanococcus maripaludis* S2 implicate formate metabolism as the target of hypophosphite inhibition. Currently, there is no known biological pathway for anaerobic hypophosphite oxidation, and hypophosphite is stable in anoxic sediments for weeks to months. Given its widespread natural occurrence, we propose that hypophosphite may modulate the carbon cycle in natural environments. Taken together, our results suggest that hypophosphite could be used as a safe, inexpensive strategy for methane control in syntrophic methanogenic ecosystems.



KEYWORDS: *methane, greenhouse gas, chemical inhibitor, reduced phosphorus, syntrophy, rice, ruminant*

INTRODUCTION

Microbial methanogenesis is responsible for over half of anthropogenic methane emissions. Much of this methane is derived from managed ecosystems where interventions are possible to minimize this flux.[1] Redirecting fermentative anaerobic systems toward increased accumulation of intermediates is recognized as an important approach to waste valorization and enhancing capture of energy in anaerobic digesters[2−4] and ruminant agriculture.[5]

Most known methanogenesis inhibitors target the conserved archaeal enzyme responsible for methane synthesis, methyl-coenzyme A reductase (McrA).[6] McrA inhibitors include bromoform (CH_3Br, the active methanogenesis inhibitor in red seaweeds),[7] 2-bromoethanesulfonic acid (BES),[8] and 3-nitroxypropanol (3-NOP).[9] These compounds are selective inhibitors of methanogenesis, but issues with stability, cost, and negative impacts on environmental systems limit their widespread use. As such, there is an unmet need for cheap, nontoxic, potent, and selective inhibitors of methanogenesis.

A large share of global microbial methane is produced as a consequence of syntrophic methanogenesis.[1] Syntrophy in methanogenic systems is the obligate, mutualistic, thermodynamic interdependence between organic acid disproportionating secondary fermentative organisms (syntrophs) that produce hydrogen, formate, and acetate and methanogens that consume these substrates.[10,11] Thus, chemical inhibitors of syntrophic bacteria or interference with acetate, hydrogen, or formate exchange represent viable targets for methane flux control. Fluoroacetate inhibits acetate metabolism in acetoclastic methanogens but is very toxic and therefore unlikely to be selective enough to be useful in natural systems.[12] Cyanide (CN) and carbon monoxide (CO) are inhibitors of hydrogenase enzymes,[13,14] and hypophosphite ($H_2PO_2^-$) is known to inhibit the bacterial molybdoenzyme formate dehydrogenase (FDH).[15,16] These compounds can inhibit some syntroph-methanogen cocultures.[17] However, their mechanisms of inhibition are poorly understood, and it remains unclear how

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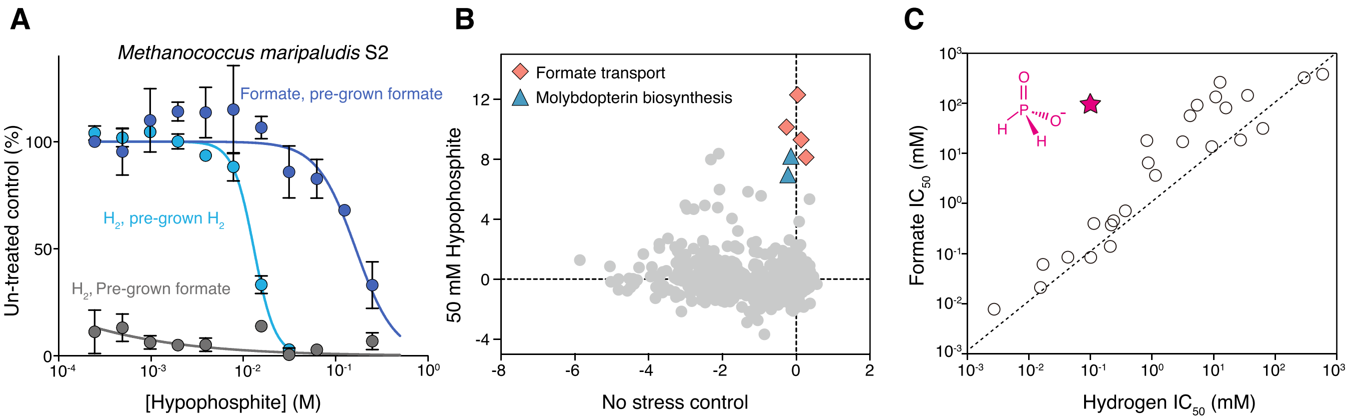

Figure 1. Formate metabolism and hypophosphite sensitivity in a model methanogen. (A) Dose−response curves showing the influence of varying concentrations of hypophosphite on *Methanococcus maripaludis* S2 growth. Different cultures were pregrown on or transferred into conditions with either hydrogen or formate as the sole electron donor. Dark blue points are cultures pregrown on formate, then transferred to formate. Light blue points are cultures pregrown on hydrogen, then transferred to hydrogen. Dark gray points are cultures pregrown on formate and then transferred to hydrogen. Points are averages, and error bars represent standard deviations of biological replicates. (B) Genome-wide gene fitness for *Methanococcus maripaludis* S2 grown on hydrogen with no inhibitor or in the presence of 50 mM hypophosphite. Colored points indicate genes involved in molybdopterin biosynthesis, MoeA (blue) and formate transport (pink). Gene fitness is the log2 ratio of the abundance of transposon insertion strains in a gene for a given condition relative to the "time-zero" strain abundance (Methods section). Beneficial, neutral, and deleterious mutations during growth in the presence of 50 mM hypophosphite are indicated by gene fitness >0, =0, and <0 on the *y*-axis, respectively. (C) Half-maximal inhibitory potencies (IC_{50}) for a set of compounds against *Methanococcus maripaludis* S2 grown with either formate or hydrogen as the sole electron donor. The red star represents sodium hypophosphite. The dashed line represents the same IC_{50} in formate- and hydrogen-grown cells.

selectively these compounds target syntrophic methanogenesis versus primary fermentation.

Both CN and CO are generally toxic to people and animals because they bind to respiratory metalloenzymes.[18,19] These compounds are also microbially degraded.[20,21] As such, CN and CO are unlikely to be useful for methane control in ruminants, rice fields, or reactors. In contrast, hypophosphite is largely nontoxic to humans and animals with an LD_{50} in rats of 1.6 g/kg.[22] Over the past 150 years, hypophosphite has been tested as a treatment for tuberculosis, a food preservative, and as a phosphorus source for animals.[22] While ineffective as a human therapeutic or animal phosphorus source, hypophosphite is used occasionally in food preservation and as a soluble counterion in bovine calcium supplements. As such, the United States Food and Drug Administration designates hypophosphite as Generally Recognized as Safe (GRAS) for use as a nutrient or dietary supplement.[22]

Given its structural similarity to formate, hypophosphite has been used as an active-site probe to study formate metabolic enzymes in fermentative bacteria. Enzyme studies show that hypophosphite is a formate-competitive, irreversible inhibitor of *Escherichia coli* formate dehydrogenase (FDH)[23] and pyruvate formate lyase (PFL).[24] The precise mechanism of irreversible FDH inhibition is not well characterized, but, in the absence of formate, submicromolar (μM) hypophosphite concentrations are sufficient to completely inactivate all FDH in *E. coli*.[15] PFL is inactivated by hypophosphite through the formation of a covalent C−P adduct at low μM concentrations.[24]

Despite the low toxicity of hypophosphite to animals and potential to interfere with microbial formate metabolism, it is unknown if hypophosphite is selectively inhibitory of syntrophs and methanogens versus primary fermentative bacteria. All studies that have used hypophosphite to study formate metabolism in methanogens and primary fermenters have used millimolar concentrations of hypophosphite[17,25] which is well above the thermodynamically favorable formate concentration thresholds for syntrophy. The one study that quantified hypophosphite inhibitory potency against a syntroph measured a low micromolar half-maximal inhibitory potency (IC_{50}) under nonsyntrophic, crotonate disproportionating conditions.[16] Studies that evaluated hypophosphite inhibition of formatotrophic methanogens report variable sensitivity and only at high millimolar concentrations.[26,27] Overall, based on evidence in the literature, it is unclear whether or not hypophosphite can effectively and selectively inhibit methane production in complex, natural methanogenic microbiomes at low micromolar (μM) concentrations.

Recent work suggests that hypophosphite is widely produced as part of a natural phosphorus redox cycle.[28] Hypophosphite is consistently measured in anaerobic wetland porewater at concentrations in the high nanomolar to low μM range.[28] Strikingly, hypophosphite concentrations can exceed 350 μM in the termite gut.[29] Some work supports a biological mechanism of phosphorus reduction in environmental systems, but the enzymes involved remain unknown.[28,30−32] Recent work from our group demonstrates that phosphite can be anaerobically oxidized coupled to methanogenesis via a lithosyntrophic association between phosphite-oxidizing bacteria and methanogens.[33] Although natural hypophosphite concentrations measured in wetlands and termite guts exceed those necessary to inhibit the *E. coli* FDH,[15] it is not known if hypophosphite can influence anaerobic carbon cycling in these systems or if it shows selectivity against formate metabolism in complex systems.

To bridge these knowledge gaps, we tested whether hypophosphite could serve as a selective methane inhibition strategy in anaerobic ecosystems. More specifically, we measured the influence of hypophosphite on a genetically tractable methanogen, on methanogenic microbiomes from rice field sediment microcosms, and on methane fluxes from rumen fluid incubations and potted rice plants. Our results implicate syntrophic formate exchange as a target of hypophosphite, and we find that hypophosphite is selectively inhibitory of syntrophic methanogenesis versus primary fermentation. We also show that hypophosphite is stable in anoxic systems and can inhibit

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methane fluxes from potted rice plants and the rumen fluid of dairy cows. Finally, comparison of the inhibitory potency of hypophosphite with measured natural concentrations in anoxic ecosystems suggests that hypophosphite could be a natural control on anaerobic carbon cycling pathways and methane fluxes. Taken together, our results have implications for methane control in agricultural ecosystems and point to important future directions of inquiry for improving our understanding of the interactions between carbon and phosphorus biogeochemical cycles.

■ RESULTS

Hypophosphite Is a Potent Inhibitor of a Model Formatotrophic/Hydrogenotrophic Methanogen

In previous work, high millimolar concentrations of hypophosphite were found to compete with formate to inhibit formatotrophic methanogens.[26,27] However, in these experiments, formate concentrations were in a similar concentration range, and subsequent work with *Methanococcus maripaludis* S2 (S2) has demonstrated that the formate metabolic pathway, including the formate transporters and FDH is regulated by the availability of formate in the culture media.[34,35] We thus hypothesized that it would be possible to measure a lower half-maximal inhibitory potency (IC_{50}) for hypophosphite by preculturing cells on formate before transferring to fresh media without formate. To test this hypothesis, we precultured S2 in a chemically defined medium with either hydrogen or formate as the sole electron donor and transferred these cultures into fresh media with varying concentrations of hypophosphite (Methods section, Supplementary Table S1). Formate-pregrown S2 transferred to fresh media with 120 mM formate as the electron donor is very insensitive to hypophosphite with an IC_{50} > 100 mM, but these same formate-pregrown cells are very sensitive to hypophosphite when transferred to cultures with hydrogen as the sole electron donor (Figure 1A). Cells pregrown on hydrogen and transferred to hydrogen have a hypophosphite IC_{50} ∼10 mM. Taken together, these results support a model in which hypophosphite is a potent inhibitor of formatotrophic methanogens only when both methanogen formate transporters are expressed and formate concentrations are low. This scenario is what we would expect to find in most natural and engineered syntrophic methanogenic systems.[10,36]

Methanogen Formate Metabolic Enzymes Are Implicated in the Mechanism of Hypophosphite Inhibition

To identify genes important and detrimental for *Methanococcus maripaludis* S2 growth in the presence of hypophosphite, we grew a pooled bar-coded transposon mutant library[37] (Methods section) of S2 with hydrogen as the sole electron donor in the presence of inhibitory concentrations of hypophosphite. Hydrogen was chosen as an electron donor for these experiments because lower concentrations of hypophosphite were inhibitory of hydrogen-grown cells (precultured on hydrogen) compared with formate-grown cells (Figure 1A). We found that mutants with transposon insertions in genes involved in formate transport and molybdopterin biosynthesis grew much better than mutants in the average gene, indicating that these genes are strongly detrimental in the presence of hypophosphite (Figure 1B).

Formate transporters are involved in the uptake of formate from the extracellular environment to the cytoplasm, and molybdopterin is a cofactor in the major formate metabolic enzymes, formylmethanofuran dehydrogenase (Fmd, MMP_RS02690-MMP_RS02710; Fwd, MMP_RS06425-MMP_RS06400) and formate dehydrogenase enzymes (FDH1, MMP_RS00795-MMP_RS00800; FDH2, MMP_RS06680-MMP_RS06685). We did not observe strong differential fitness during growth on hypophosphite for any of these major formate metabolic enzymes, but this may be because the multiple paralogues of these formate enzymes are potentially redundant. We were able to confirm the sensitivity of individual strains lacking two MoeA genes (MoeA, MMP_RS08320; MoeA3, MMP_RS02885) (Figure S1). MoeA is the enzyme responsible for the insertion of molybdenum into the cofactor in formate dehydrogenase and other molybdopterin enzymes.

There are several possible explanations for the fitness advantage of molybdopterin biosynthesis enzyme mutants in resistance to hypophosphite. Hypophosphite may influence regulation of molybdopterin biosynthesis and/or directly react with molybdopterin cofactor or molybdopterin enzymes. If hypophosphite is oxidized by oxidized molybdopterin, the product would likely be phosphite or phosphate, whereas if hypophosphite is reduced, the product would be phosphine. Conceivably, either of these reactions could be detrimental for fitness as hypophosphite oxidation would disrupt electron transfer, while hypophosphite reduction would produce the toxic gas phosphine which could react with metalloenzymes in the cell. While the precise chemical mechanism of potent irreversible inhibition of FDH by hypophosphite is unknown,[23] it likely involves a redox reaction. This is likely not the case for formate transporters which do not have redox cofactors and are unlikely, therefore, to react with hypophosphite. Regardless, our results implicate methanogen formate metabolism as part of the mechanism of hypophosphite inhibition.

Hypophosphite Is a Uniquely Selective Inhibitor of Formatotrophic Methanogenesis

To determine if other compounds are similar to hypophosphite in their selective inhibition of formate metabolism in *Methanococcus maripaludis* S2, we measured the inhibitory potency of a panel of inorganic ions against S2 cultured under either hydrogenotrophic or formatotrophic growth conditions. For these assays, S2 was pregrown on formate, then transferred into either formate or hydrogen media and mixed with serially diluted compounds in 384-well microplates. By measuring growth relative to uninhibited control cultures, we determined the IC_{50} values of diverse compounds. Most compounds have similar IC_{50}s between hydrogen- and formate-grown conditions, but hypophosphite is much less inhibitory for formate-grown cells. (Figure 1C and Supporting Information Table S2). This result is consistent with hypophosphite being competitive with that of formate. High concentrations of formate present in formatotrophic media prevent hypophosphite from entering cells and inhibiting enzymes. Taken together, our findings support the view that hypophosphite is a uniquely selective inhibitor of methanogen formate metabolism.

Hypophosphite Is a Selective Inhibitor of Methanogenesis in Fermentative, Anaerobic Microcosms from Rice Field Sediment

Based on the potent inhibition of*Methanococcus*, we next examined whether hypophosphite can selectively suppress methane production in a complex microbiome. We amended anoxic chemically defined microbial growth media with rice field sediment and a complex fermentable electron donor/carbon source in the form of 1 g/L yeast extract (Methods section, Supplementary Table S3). We passaged these microcosms three

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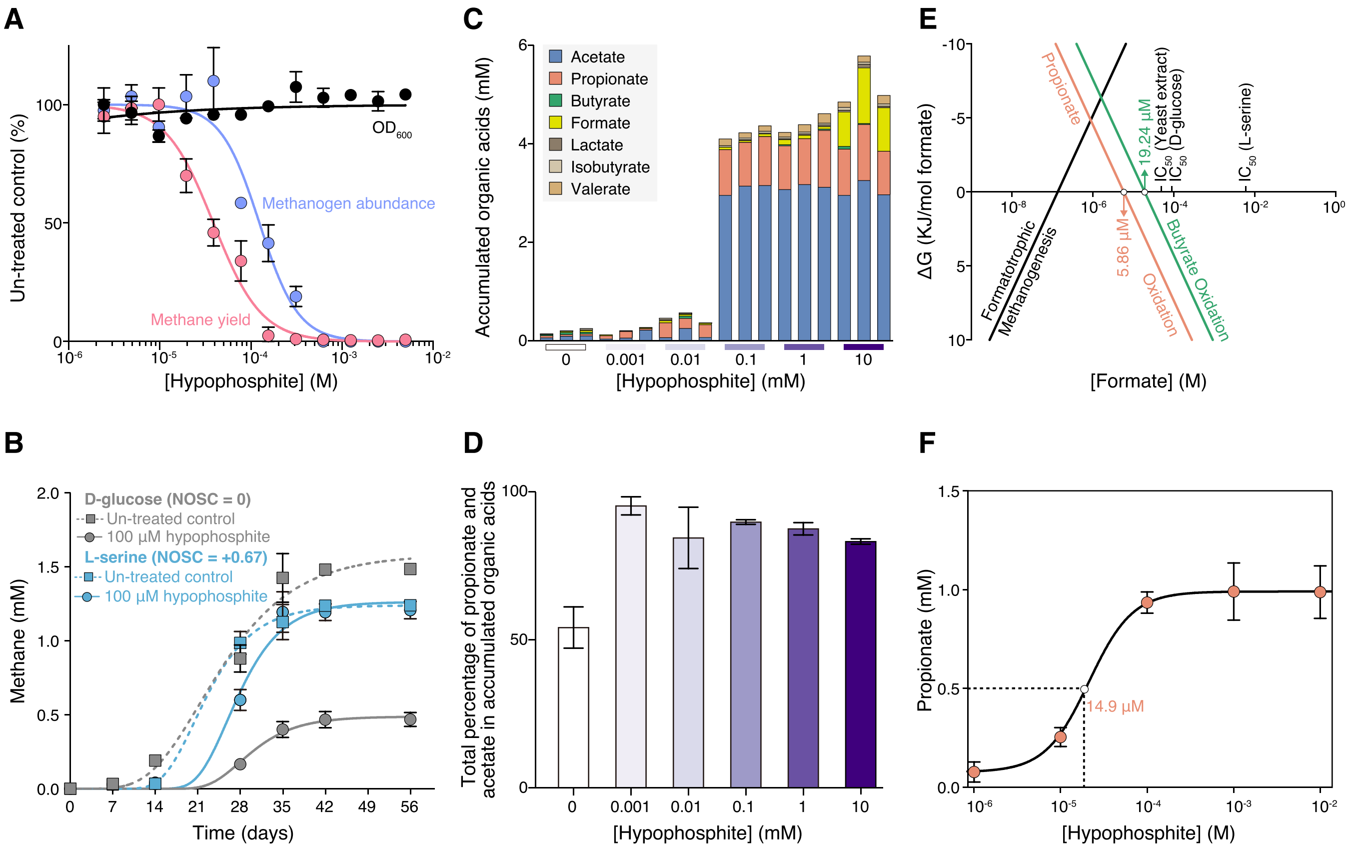

Figure 2. Inhibition of methanogenesis by hypophosphite in syntrophic methanogenic microcosms from rice fields. (A) Dose−response curves showing the influence of varying concentrations of hypophosphite on methane production, growth yield (OD600), and the relative abundance of methanogenic archaea (via 16S rDNA amplicon sequencing) in fermentative methanogenic rice field microcosms with yeast extract as the sole electron donor. Points are averages and error bars represent standard deviations of biological replicates. (B) Hypophosphite inhibition of methane production in replicate microcosms with D-glucose (light gray) or L-serine (sky-blue) as the sole electron donor. (C) Organic acids after 28 days in D-glucose microcosms with varying concentrations of hypophosphite Points are averages and error bars represent standard deviations of biological replicates. (D) Propionate and acetate as a proportion of total organic acid electrons in D-glucose microcosms. (E) Overall Gibbs energy (ΔG_r, kJ/mol formate) of formatotrophic methanogenesis or syntrophic butyrate or propionate oxidation (37 °C) as a function of formate concentrations compared with IC_{50}s against methane yields in rice field microcosms. (F) Propionate concentrations measured in methanogenic D-glucose microcosms as a function of hypophosphite concentrations compared with the free energy of syntrophic propionate oxidation as a function of formate concentrations. Half-maximal concentrations of propionate accumulation are shown with dotted lines. Points are averages, and error bars represent standard deviations of biological replicates.

times to dilute away sediment until we could measure growth by monitoring optical density (OD600) and then transferred these enriched microcosms into fresh cultures containing varying concentrations of sodium hypophosphite. We monitored methane, growth and methanogen abundance using 16S rDNA amplicons at the early stationary phase of methane production in control and treated cultures after 3 weeks. We found that methane and the relative abundance of methanogens were inhibited with an IC_{50} of ∼50 μM hypophosphite, but optical density, indicative of fermentative growth, was not inhibited by at least 10 mM sodium hypophosphite (Figure 2A). This result shows that hypophosphite can selectively inhibit methanogenesis versus primary fermentation in a complex fermentative anaerobic microbiome.

Hypophosphite Is More Inhibitory of Methanogenic Systems that Rely More Heavily on Syntrophic Electron Exchange

Syntrophic organic acid oxidation to formate becomes thermodynamically unfavorable above concentrations of ∼10 μM formate.[36] Thus, we hypothesized that interference with formate electron exchange is an important mechanism of hypophosphite inhibition. We also hypothesized that the degree to which a methanogenic system relies on syntrophic electron exchange influences hypophosphite sensitivity.

To evaluate the importance of syntrophy in controlling hypophosphite inhibitory potency, we prepared rice field sediment microcosms with electron donors that vary in nominal oxidation state of organic carbon (NOSC). We previously demonstrated that fermentative methanogenic degradation of more reduced organic carbon (negative or zero NOSC) relies more heavily on syntrophic methanogenesis compared to degradation of more oxidized organic carbon.[38] Consistent with our hypothesis, we found that hypophosphite is a more potent inhibitor of methanogenesis in microcosms amended with D-glucose (NOSC = 0) compared to microcosms amended with L-serine (NOSC = +0.67). After 56 h of incubation with 100 μM hypophosphite, methane production in D-glucose-amended microcosms was reduced to ∼25% of the untreated controls, whereas methane production in L-serine-amended microcosms was less affected (Figure 2B). Notably, D-glucose is a dominant electron donor in rice field sediments[39] and ruminants and syntrophy is important in both of these systems.

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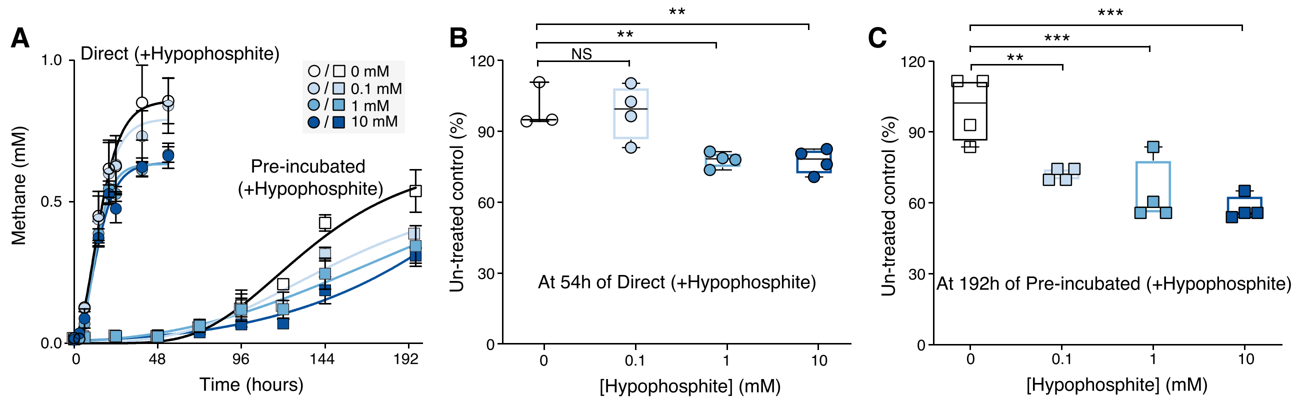

Figure 3. Hypophosphite inhibition of methane production in bovine rumen fluid. (A) Methane production over time in incubated rumen fluid incubations amended with varying concentrations of hypophosphite across 0 to 10 mM. Circles are incubations where rumen fluid is added directly to a buffer containing hypophosphite. Squares are preincubated rumen fluid transferred into incubations with varying concentrations of hypophosphite. Points are averages and error bars represent standard deviations of biological replicates. (B and C) Methane production relative to untreated controls (%) for 0.1, 1, and 10 mM hypophosphite-treated incubations (B) Direct incubations, and (C) Preincubated enrichments. Statistics analysis of significance was performed by ANOVA, $P < 0.001$ (***); $P < 0.01$ (**).

Hypophosphite Likely Competes with Formate Exchange in Syntrophic Methanogenesis

To further evaluate the competitive interference of hypophosphite with syntrophic formatotrophic methanogenesis, we measured organic acid concentrations after 28 days in D-glucose and L-serine cultures in the presence of varying concentrations of hypophosphite. We observed accumulation of acetate, propionate, butyrate, and other minor organic acids in D-glucose cultures amended with 100 μM or more hypophosphite (Figure 2C, Supplemental Table S4). We also observed organic acid accumulation at 1 mM hypophosphite in L-serine cultures (Supplemental Table S5). These results are further evidence of inhibition of syntrophic methanogenesis while primary fermentation proceeds. Additionally, we found that propionate and acetate accumulate in D-glucose cultures relative to other organic acids at concentrations of 1 μM hypophosphite and above (Figure 2D). In diverse terrestrial wetlands, hypophosphite concentrations can reach 2 μM,[28] and therefore, our observation suggests that naturally produced hypophosphite may influence carbon flow in methanogenic microbiomes.

We next compared measured hypophosphite IC_{50} values against the thermodynamically favorable formate concentration thresholds for syntrophic butyrate and propionate oxidation. We found that IC_{50}s for the D-glucose and yeast extract cultures, which rely more heavily on syntrophic electron exchange,[38] are close to the thermodynamic formate concentration thresholds (Figure 2E). Direct quantification of formate at low micromolar concentrations is technically challenging. We instead examined propionate accumulation in D-glucose cultures amended with varying concentrations of hypophosphite (1 μM−10 mM). We found that the concentrations of hypophosphite equal to or above 10 μM (EC_{50} = 14.9 μM) were sufficient to drive propionate accumulation (Figure 2F). This result is consistent with a model where hypophosphite interferes with formate exchange in syntrophic methanogenesis and thereby stalls syntrophic propionate oxidation.

Hypophosphite Is Not a Potent Inhibitor of Other Respiratory Metabolisms or Methane Oxidation

To further assess the selectivity of hypophosphite against syntrophic methanogenesis versus other metabolisms, we prepared rice field sediment microcosms with 1 g/L yeast extract as the sole electron donor and added sulfate or nitrate as terminal electron acceptors. Again, after passaging these cultures multiple times to remove residual sediment that interferes with measurements of optical density, we measured growth by optical density and the accumulation of nitrite or sulfide using colorimetric assays. We found that neither fermentative growth nor nitrate or sulfate reduction was inhibited in these cultures up to a concentration of 10 mM sodium hypophosphite (Supporting Information Figure S2A). We also measured the inhibitory potency against a model aerobic methanotroph, *Methylosinus trichosporium* OB3B, grown with methane as the sole electron donor, and against a model aerobic heterotroph, *E. coli* K-12. We found that these organisms had IC_{50}s above 10 mM hypophosphite (Supporting Information Figure S2B). Along with the results presented in Figures 1 and 2, these observations further support the selective inhibitory effect of syntrophic methanogenesis by hypophosphite relative to primary fermentation or other respiratory metabolisms and are consistent with other work showing that growth of diverse bacteria is not greatly impacted by concentrations of hypophosphite up to 10 mM.[25,40]

Hypophosphite Inhibits Methane Production in Rumen Fluid from Cows

Ruminants are responsible for around 16% of global methane fluxes.[41] Syntrophy is important in the rumen. In fact, the first “S organism” syntroph was derived from a methanogenic rumen culture.[42] As such, there is widespread interest in developing strategies for methane mitigation for ruminants. We prepared anoxic rumen enrichments in serum bottles by mixing rumen fluid with a concentrated bicarbonate buffer to neutralize organic acids and amended with finely milled cattle feed as the sole electron donor and varying concentrations of hypophosphite (Methods section). We found that 1 mM hypophosphite inhibited ∼25% of total methane production after 24 h of incubation at 37 °C, but higher doses did not lead to significantly more inhibition (Figure 3A,B). This is consistent with reports that formate is responsible for ∼30% of the total methane flux from the bovine rumen.[43] We next preincubated the rumen fluid in the absence of hypophosphite for 7 days to allow for consumption of syntrophic intermediates that may have accumulated such that syntrophs and methanogens were thermodynamically uncoupled. In this preincubated rumen fluid, we found that 0.1 mM hypophosphite was capable of

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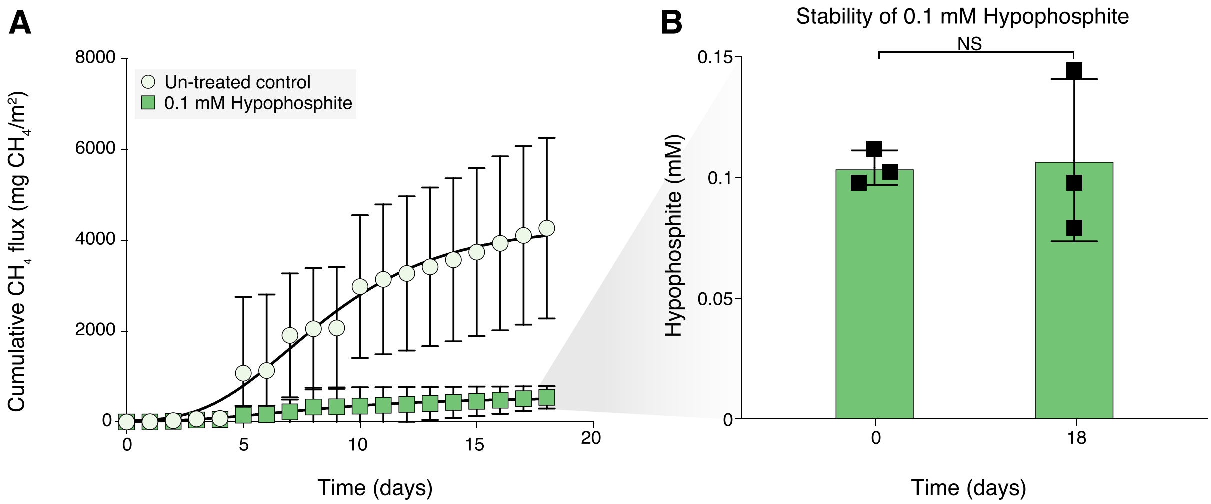

Figure 4. Hypophosphite inhibition of methane fluxes in potted rice plants. (A) Cumulative methane in untreated control rice pots ($n = 3$) compared with 0.1 mM hypophosphite treated rice pots ($n = 3$). Points are averages and error bars represent standard deviations of biological replicates. (B) Hypophosphite concentrations at initial time point and after 18 days in rice pot porewater. Statistics analysis of significance were performed by t tests, $P < 0.05$ (*).

inhibition of 25% of total methane after 192 h (Figure 3A,C). This result is consistent with inhibition of formate exchange in syntrophic methanogenesis (Figures 1 and 2), but future work measuring methane in cattle treated with hypophosphite is necessary to determine the efficacy in cattle methane mitigation.

Hypophosphite Inhibits Methane Production in Potted Rice Plants and Remains Stable in Anaerobic Sediments

Alongside ruminants, rice agriculture is another major global source of methane.[1] To evaluate the potential of hypophosphite to inhibit methanogenesis in rice cultivation systems, we treated flooded rice pots with 0.1 mM hypophosphite and monitored methane over time (Methods section). We found that methane production was inhibited up to 80% by hypophosphite after 18 days (Figure 4A). No impact on the plant health or rice yield was observed. We also monitored the stability of hypophosphite in rice pot porewater and measured no significant loss after 18 days (Figure 4B). Currently, no anaerobic microbial pathways of hypophosphite oxidation are known. However, hypophosphite is prone to abiotic oxidation in the presence of metals and oxygen and has a short half-life in oxidized sediments.[28] Notably, the oxidation of hypophosphite would result in phosphate as the major end product, thereby converting this methanogenesis inhibitor into an essential nutrient for plants.

■ DISCUSSION

Here, we provide evidence that hypophosphite, an inorganic formate analogue, is a potent and selective inhibitor of formate exchange during syntrophic methanogenesis. Physiological and genetic experiments implicate formate metabolic enzymes in a model methanogen as targets of hypophosphite. In complex fermentative methanogenic microbiomes, hypophosphite concentrations that exceed the thermodynamically favorable formate concentration range for syntrophy inhibit methane production and lead to the accumulation of organic acids while allowing primary fermentation to proceed. Preliminary tests in complex potted rice plant mesocosms and rumen fluid from dairy cows suggest that hypophosphite could be useful for modulating methane fluxes in diverse fermentative methanogenic ecosystems, where syntrophic electron exchange is important.

The apparent selectivity and potency of hypophosphite as an inhibitor of formatotrophic methanogenesis suggest that it could be a useful tool for studying the relative importance of formate as an electron carrier in diverse syntrophic methanogenic systems. While formate diffuses ~5× more slowly than hydrogen, formate is 250-fold higher in concentration when in thermodynamic equilibrium with hydrogen.[10,36] Thus, it is thought that hydrogen is more important as an aqueous electron carrier over longer distances, while formate is dominant as an electron shuttle in biofilms and methanogenic microbial aggregates such as anaerobic digester granules.[36]

Hypophosphite is an inexpensive, inorganic compound and is stable in anoxic ecosystems. In potted rice plants and rumen fluid from cows, 100 μM to 1 mM hypophosphite inhibits methanogenesis to 25−75% of controls. At these concentrations, hypophosphite is nontoxic to rice plants or ruminants. Because hypophosphite targets formate utilization, it is not expected to eliminate methane production entirely in systems where methanogens can use other electron carriers (e.g., hydrogen). This is consistent with our results in cow rumen fluid where ~25−30% of methane production is suppressible and it is thought that a similar proportion of methane is derived from formate.[43] Future work will explore the potential benefits for methanogenesis inhibition from combining hypophosphite with inhibitors of hydrogenotrophic methanogenesis.

Also, future research will focus on determining whether hypophosphite can function as both a methanogenesis inhibitor and phosphorus source in rice fields and as a methane control strategy in ruminant agriculture. In the course of this work, the safety and impact on animal, human, and plant health of hypophosphite amendments will be carefully evaluated. Evaluating hypophosphite in other industrial ecosystems where methane control is important, such as anaerobic digesters, manure lagoons, or subsurface hydrogen gas storage, is also a future area of investigation. In forthcoming work, we have evidence that hypophosphite is effective at controlling methane in laboratory-scale anaerobic digester systems (Matt Weaver et al., unpublished results).

Additionally, our results suggest that naturally produced hypophosphite could modulate anaerobic carbon cycling and

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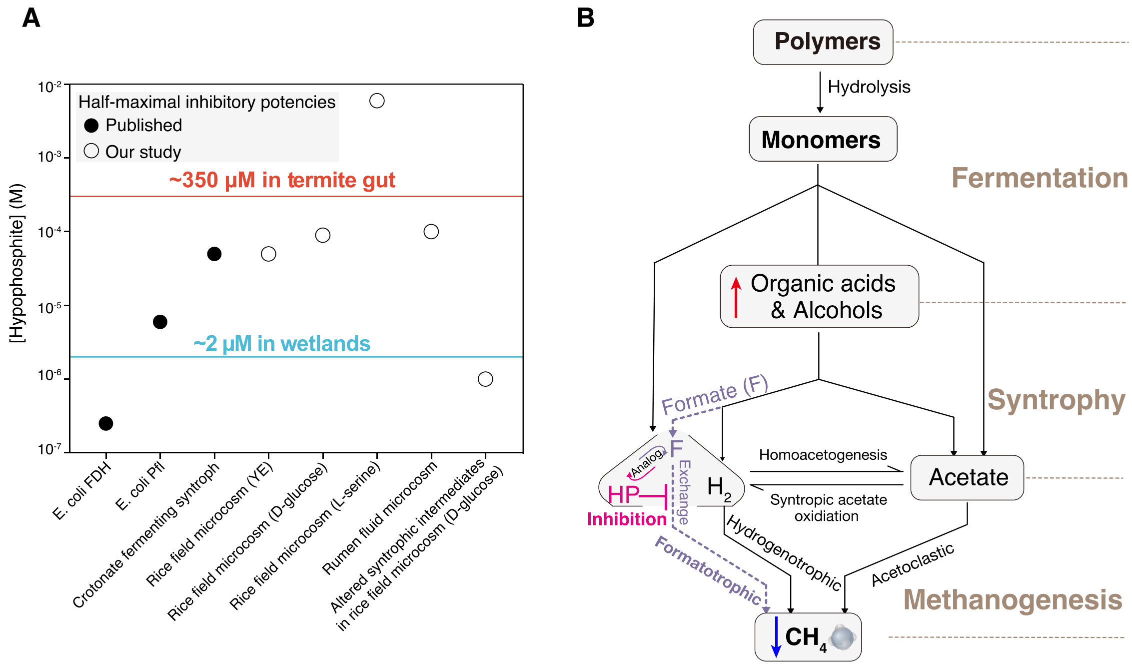

Figure 5. Hypophosphite as a natural, potent, and selective inhibitor of formate exchange in syntrophic methanogenesis. (A) Half-maximal inhibitory potencies of hypophosphite against different microbial enzymes or microbial processes compared with environmental concentrations. (B) Conceptual model showing likely targets of hypophosphite in complex carbon and electron flow processes in fermentative methanogenic systems.

methane fluxes in anoxic systems such as wetlands or the termite gut. Phosphorus undergoes a natural redox cycle.[28] A survey of anoxic aquatic sediments found that reduced phosphorus represented up to a third of total phosphorus,[28] with hypophosphite concentrations up to ∼2 μM. In termite guts, hypophosphite was measured at >300 μM.[29] While the sources of reduced phosphorus in these environments are unknown, it is possible that hypophosphite could be produced through biological phosphorus reduction.[28,30−32,44]

We compared the measured inhibitory potency of hypophosphite against diverse formate metabolic enzymes and microbial processes with natural concentrations in different ecosystems (Figure 5A). It is clear from this analysis that hypophosphite concentrations in aquatic sediments are sufficient to inhibit microbial formate dehydrogenase enzymes and cause a shift in syntrophic organic acid composition (as shown in Figures 1 and 2). In termite guts, hypophosphite concentrations exceed the inhibitory concentrations required to inhibit syntrophic methanogenesis in our microcosms from rice fields and cow rumens. Termite guts, unlike the rumen or most other metazoan guts, have a dominance of homoacetogenic bacteria over methanogenic archaea, and most methane is derived from hydrogen and acetate rather than formate in termite guts.[45] The basis for homoacetogenesis in termite guts has long been debated, but based on our results, we hypothesize that high hypophosphite concentrations are a plausible explanation that to our knowledge has not been discussed in the literature. By annotating formate dehydrogenase and formate transporter genes across 893 archaeal methanogen genomes from the Genome Taxonomy Database (GTDB),[46] we found that more than 30% of methanogens possess the genetic potential for formate utilization (Supplementary Figure S3). This prevalence indicates that formatotrophy is widespread among methanogenic archaea and represents a phylogenetically broad potential target for hypophosphite inhibition.

We propose that hypophosphite selectively inhibits syntrophic methanogenesis by competing with formate for uptake by formatotrophic methanogens (Figure 5B) while allowing primary fermentation to proceed. While higher concentrations of hypophosphite can interfere with fermentation and respiration, the selectivity window is several orders of magnitude. Identifying and characterizing the enzymes responsible for phosphorus reduction is critical for understanding the mechanisms that regulate steady-state concentrations of reduced phosphorus and its role in modulating methane flux. Understanding the interactions between the anaerobic carbon and phosphorus cycle has implications for biogeochemical modeling of these cycles.

Environmental Implication

Most strategies for microbial methane control involve toxic, expensive, and rapidly degraded organic compounds. Hypophosphite is an inorganic compound with a long half-life in the absence of oxygen. Here, we provide the first direct evidence that hypophosphite can selectively interfere with syntrophic methanogenesis in rice fields and ruminants but is relatively nontoxic to primary fermentative bacteria, respiratory bacteria, plants, and animals. We also propose that naturally produced hypophosphite can modulate methane fluxes in natural anaerobic sediments and termite guts, suggesting a previously overlooked interaction between the carbon and phosphorus redox cycles. This work identifies an abundant natural compound that safely suppresses methane production in anaerobic environments by targeting microbial interactions rather than killing the microbes or blocking essential metabolic enzymes in methanogens.

■ METHODS

Media and Cultivation Conditions

All medium recipes are in Supporting Information. *Methanococcus maripaludis* S2 was cultivated in basal media in anaerobic Balch tubes as

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described previously[37] or in microplates in an anoxic pressurized incubation vessel using either formate or hydrogen as the electron donor. *E. coli* was cultured in LB broth (Difco) and in NMS media in sealed Balch tubes with a headspace of CH_4 and air. Rice field microcosms were cultured in a chemically defined basal medium as described previously.[38] Organic carbon compounds, stressors, and sodium hypophosphite were added from concentrated anoxic stock solutions (Sigma-Aldrich). To assess the inhibitory potency of different concentrations of hypophosphite against various microorganisms or microbial processes, concentrated hypophosphite stocks were diluted into the culture media.

Analytical Methods

Headspace methane concentrations in anaerobic Balch tubes and serum bottle cultures were measured with an Agilent gas chromatograph model 7890A (G344OA) with a Superlco SP-2380 Fused Silica Capillary column containing a proprietary, bonded silica-based stationary phase. Methane was measured using a flame ionization detector (FID) and GC oven temperature was held at 50 °C for 60 min to allow for noncarrier gases to elute. The FID heater was kept at 300 °C; air flow was 400 mL/min; hydrogen gas fuel flow was 30 mL/min; and nitrogen gas flow was 25 mL/min. (H_2 and N_2 are the carrier gases for FID). Methane standard gases (Scotty Analyzed Gases) were used for calibration. GC ChemStation (Agilent Technologies) computer software was used for quantification. Growth of anaerobic Balch tube and serum bottle cultures was measured using a Spectronic 20D spectrophotometer. Hypophosphite concentrations were measured using ion chromatography (Dionex).[47] To measure the syntrophic intermediate metabolites across anaerobic Balch tubes treated with glucose, we quantified formate, acetate, lactate, propionate, butyrate, isobutyrate, and valerate by high-performance liquid chromatography on a Shimadzu LC-20AD using an Aminex HPC-87H, 300 mm × 7.8 mm with a 5 mM sulfuric acid mobile phase at a flow rate of 0.6 mL/min. The pressure and temperature were held at 720 psi and 50 °C. Standard curves were prepared for quantification of each organic acid.

Dose−Response Assays and Analysis with *M. maripaludis* S2

For high-throughput dose−response assays concentrated stock solutions were serially Z-diluted across three 384-well plates to attain 12 2-fold dilutions of each compound.[48−50] These arrayed dose−response plates were inoculated 1:1 with 2× concentrated basal media with *Methanococcus*. For hydrogen growth assays, plates were sealed with breathable seals and grown at 37 °C in pressure vessels with a pressure of 30 psi 80:20 H_2:CO_2 atmosphere. For formate growth assays, plates were sealed in permeable seals and grown in anaerobic boxes (BD). All growth vessels contained paper-towels soaked in 1 M sodium sulfide as a sulfur source and atmospheric reductant. At time points, vessels were opened inside anaerobic chambers (Coy) and OD600 measured with a Tecan M1000 Pro microplate reader (Tecan). Plate reads were uploaded to an in-house growth database. Early stationary phase time points were selected for dose−response analysis and custom scripts to fit a nonlinear regression to determine half-maximal inhibitory concentrations (IC_{50}).[50]

Genome-Wide Fitness Assays to Identify Genes Important for Hypophosphite Resistance and Sensitivity in *M. maripaludis* S2

A previously constructed bar-coded transposon mutant library of *M. maripaludis* S2[37] was recovered from freezer stocks and used as previously described. All media recipes are in the Supporting Information. Hypophosphite was added from concentrated aqueous stock solutions for dose−response assays. Anaerobic Balch tubes were incubated at 37 °C, and growth was monitored over time by OD600. Cultures that were inhibited by hypophosphite relative to untreated controls were allowed to grow to the stationary phase before collecting pellets for measuring genome-wide fitness using random-barcode transposon sequencing (RB-TnSeq). We represent strain fitness as the normalized log2 ratio of barcode abundance between the treatment sample (i.e., after growth in a certain condition) and the reference "time-zero" samples.[51,52] Gene fitness is the weighted average of the strain fitness (Supplementary Table S6). Gene fitness for these experiments is available on fit.genomics.lbl.gov as set15S713, set15S714, set15S711, set15S712.

Dose−Response Assays and Analysis of Rice Field Microcosms

Rice field microcosms were cultured in a chemically defined basal medium as described previously in Balch tubes.[38] Organic carbon compounds, stressors, and sodium hypophosphite were added from concentrated anoxic stock solutions (Sigma-Aldrich). Anoxic sodium hypophosphite stock solutions in water were serially diluted in Balch tubes prior to amendment with enriched microcosms. Growth was monitored by OD600 using a Spectronic 20D spectrophotometer. Methane and organic acids were measured using methods described in the Analytical Methods section.

Thermodynamic Calculations

The Gibbs energy of reaction, ΔG_r, was calculated for formatotrophic methanogenesis (eq R1), butyrate oxidation to formate coupled to bicarbonate reduction to acetate (eq R2), and propionate oxidation to formate coupled to bicarbonate reduction to acetate (eq R3) at 37 °C, as described in our previous work.[33] pH was set at 7, bicarbonate concentration was set at 2.98×10^{-2} M, acetate concentration was set at 8.80×10^{-5} M, methane concentration was set at 1×10^{-4} M, butyrate concentration was set at 3.5×10^{-5} M, and propionate concentration was set at 3.8×10^{-5} M. Formate concentration ranged from 1×10^{-10} to 1 M.

$$H_2CO_2^- + H_2O + H^+ \rightarrow HCO_3^- + CH_4(g) \qquad (R1)$$

$$2C_4H_7O_2^- + 6HCO_3^- \rightarrow 8HCO_2^- + 3C_2H_3O_2^- + 3H^+ \qquad (R2)$$

$$C_3H_5O_2^- + 2HCO_3^- \rightarrow 3HCO_2^- + C_2H_3O_2^- + H^+ \qquad (R3)$$

Generation of *M. maripaludis* moeA3 Deletion Strain

The moeA1 deletion strain was generated in a previous publication.[53] To create the *M. maripaludis* moeA3 deletion strain, two PCR fragments were initially generated. Briefly, 1000 bp upstream and 1000 bp downstream of the moeA3 gene were PCR amplified. The DNA fragments were then assembled into XbaI and NotI-digested pCRuptneo plasmid via Gibson assembly and transformed into *E. coli* DH5 via electroporation. Colonies grown on lysogeny broth solidified with 1.5% agar and ampicillin were screened by PCR to identify those carrying the correct recombinant plasmid construct. The plasmids used in this study were transformed into *M. maripaludis* S2 under anoxic conditions using a polyethylene glycol (PEG)-mediated transformation protocol. Briefly, *M. maripaludis* cultures grew to late log phase with an OD600 of ∼0.9. The cell pellet was harvested by centrifugation and washed with a transformation buffer (TB) (50 mM Tris, 350 mM sucrose, 380 mM NaCl, 1 mM MgCl2, pH 7.5). The washed cell pellet was resuspended in 0.375 mL of TB followed by the addition of ∼5 μg of plasmid DNA. Next, 0.225 mL of PEG solution (40 wt %/vol PEG 8000 in TB) was added to the cell mix, followed by a 1 h nonshaking incubation at 37 °C. Transformants were washed with fresh McCas medium and pressurized to 280 kPa with an H_2:CO_2 (80%:20%) gas mix. After outgrowing at 37 °C with agitation for at least 4 h, transformants were subinoculated into McCas medium containing neomycin (1 mg mL^{-1}) to select for transformants. For counter-selection of the recombinant suicide vector pCRuptneo, the transformants were plated on solid McCas medium containing 6-azauracil (0.2 g L^{-1}) followed by incubation in a pressurized container with 138 kPa with a H_2:CO_2 (80%:20%) gas mix at 37 °C. The *M. maripaludis* colonies were screened for the deletion of moeA3 using the primers that bind ∼500 bp upstream or downstream of the *moeA3* gene.

Methane Inhibition Assays in Rumen Fluid Incubations

Rumen fluid for in vitro rumen fermentation was obtained from three rumen-fistulated nonlactating Holstein cows housed at the University of California, Davis (UCD) Dairy Teaching and Research Facility. The donor animals were fed a dry cow total mixed ration (TMR) consisting of 50% wheat, 25% alfalfa, 21.43% almond hulls, and 3.57% dry cow pellets on a dry matter (DM) basis. Animals were fed four times a day at 5:00 am, 11:00 am, 6:00 pm, and 11:00 pm and had free access to water. Rumen fluid was collected 1 h before feeding and just prior to the start of the in vitro rumen fermentation, in accordance with the Institution of Animal Care and Use Committee (IACUC) at UCD under protocol number 22753. Approximately 500 mL of rumen fluid was collected from each cow via a perforated PVC pipe, 500 mL syringe, and Tygon tubing (Saint-Gobain North America, Malvern, PA, USA), passed through a strainer, transferred into individually prewarmed insulated thermos (Coleman Company, Chicago, IL, USA), and transported to the laboratory within 30 min. Equal volumes of rumen fluid from the three cows were pooled to account for biological variation, filtered through 2 layers of cheesecloth, dispensed into shipment containers, and flushed with CO_2 for 30 min before sealing.

Rumen fluid was stored in insulated shipment thermos bottles until initiation of the experiment (3 h). Rumen fluid was mixed with concentrated bicarbonate buffer[7] and aliquoted into Balch tubes containing dry cow total mixed ration (TMR). Hypophosphite was added from concentrated aqueous stock solutions for dose−response assays. Anaerobic Balch tubes were incubated at 37 °C and methane measured over time using gas chromatography. ANOVA analysis (a significance level set at $P < 0.05$ *, $P < 0.01$ **, $P < 0.001$ ***) was used to explore the differences of hypophosphite concentrations between untreated and 0.1, 1, and 10 mM hypophosphite-treated Rumen fluids.

Rice Pot Methane Inhibition Assays

Rice pot experiments were performed in a glasshouse at the International Rice Research Center. Two-week-old seedlings of IRRI 154 were transplanted into pots in a randomized, complete block design. Pots were maintained in a water tank in a continuously flooded manner. Fertilizers were added at rates of 130 kg N/ha, 50 kg P_2O_5/ha, and 30 kg K_2O/ha. After 21 days of cultivation, sodium hypophosphite was applied at a rate of 0.1 mM/kg dry soil. Methane was monitored over time with a Licor-3100 using a flux box that was placed over each pot daily between 7 and 11 am for 3 min per pot. Porewater was collected for measurement of hypophosphite concentrations at 0 and 18 days of treatment and analyzed by ion chromatography, as described above. Aboveground biomass (green leaves, dead leaves, stems, and panicles) was measured at the end of the experiment. No significant difference in yields and health was observed between untreated and 0.1 mM hypophosphite-treated rice plants. *t* tests (a significance level set at $P < 0.05$) were used to explore the differences of hypophosphite concentrations between untreated and 0.1 mM hypophosphite-treated rice plants.

Metabolic Annotations for Formate Utilization across Archaeal Methanogens from the GTDB Database

To investigate the prevalence of formate utilization across diverse archaeal methanogens, we first filtered the high- or medium-quality (completeness ≥50% and contamination ≤10%) archaeal genomes from GDTB database[46] by searching for methyl-coenzyme M reductase genes (KO ids: K00399, K00401, and K00402). A total of 893 high- or medium-quality archaeal genomes across 48 families were identified as methanogens and collected. Based on KO ids for formate dehydrogenase (K00125 and K22516) and formate transporter (K21993), we used the DRAM v1.5.0[54] to predict the formate metabolic genes across 893 methanogenic archaea genomes (Supplemental Table S7). The phylogenomic tree of these methanogenic archaea was constructed by GToTree v1.8.4[55] with default setting parameters.

■ ASSOCIATED CONTENT

Data Availability Statement

All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supporting Information. The nucleotide sequences for 16S rDNA gene amplicon sequencing were deposited in the SRA database under accession numbers PRJNA1188626. RB-TnSeq data is available on fit.genomics.lbl.gov. All software packages utilized in this study are publicly accessible, and no original code is reported in this study.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.6c03870.

Chemically defined basal media (Table S1); half-maximal inhibitory potencies (Table S2); chemically defined basal media for rice field microcosm experiments (Table S3); organic acid concentrations after 28 days (Tables S4 and S5); genome-wide fitness data (Table S6); and metabolic annotations for formate utilization (Table S7) (XLSX)

Mutants in *Methanococcus maripaludis* S2 MoeA genes (Figure S1); dose-response curves (Figure S2); and metabolic annotations for formate utilization (Figure S3) (PDF)

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H.K.C., R.H., M.E.W., L.A.D., and J.M.M. performed the laboratory work. R.H., H.S.A., M.N.P., L.A.D., and H.K.C. carried out the bioinformatics and statistical analysis. R.H. and H.K.C. wrote the first draft of the manuscript. All authors discussed the results. All authors read and approved the final version of the manuscript.

Notes

The authors declare the following competing financial interest(s): The corresponding author, Hans Carlson, declares he holds IP in the use of hypophosphite for control of methanogenesis. The other authors declare no competing financial interests.

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